

CSE 106

Online on Graphs (C1/C2)

In a software project, there are **N** modules, numbered from **1** to **N** . Certain modules depend on the completion of others before they can begin, represented by directed edges. An edge from module **A** to module **B** means module **A** must be compiled before module **B** starts compiling.

Your task is to determine if there is a circular dependency (cycle) in the project dependencies. A cycle would indicate a deadlock, making it impossible to complete all modules.

Input

- The first line contains two integers **N** and **M**, the number of modules and the number of dependencies.
- The next **M** lines each contain two integers **A** and **B**, indicating that module **A** must be compiled before module **B**.

Output

Print "Yes" if there is a cycle (circular dependency) in the project, and "No" otherwise.

Input	Output
4 4 1 2 2 3 3 4 4 2	Yes
5 3 1 2 2 3 4 5	No